

**EXPERIMENT NO. 6**

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**TITLE**

Multiple Linear Regression – House Price from Area and Age

**OBJECTIVE**

To predict house selling price from area and age using multiple linear regression with the normal equation, and evaluate it with MSE and R-squared with a 3D plot.

**AIM**

To fit  $\hat{Y} = b_0 + b_1X_1 + b_2X_2$  on the 6-house data, predict a new house price, and assess the fit.

**THEORY**

Regression plane (two variables):

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2$$

Normal equation with bias column  $X = [\mathbf{1} \ X_1 \ X_2]$ :

$$\theta = (X^T X)^{-1} X^T y, \quad \theta = [b_0, b_1, b_2]^T$$

Prediction and goodness of fit:

$$\hat{Y}_{new} = b_0 + b_1X_{1,new} + b_2X_{2,new} \quad \text{MSE} = \frac{1}{n} \sum (Y_i - \hat{Y}_i)^2 \quad R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

**PROCEDURE**

1. Store Area ( $X_1$ ), Age ( $X_2$ ) and Price ( $Y$ ) of 6 houses in NumPy arrays.
2. Build  $X$  with a bias column and solve  $\theta = (X^T X)^{-1} X^T y$ .
3. Predict the price of a new house (Area 2.8, Age 4).
4. Compute MSE and  $R^2$ ; plot houses with the fitted plane in 3D.

**DATASET DESCRIPTION**

Dataset Name: House price vs area and age (given table)

Source: 6 houses;  $X_1$  = area,  $X_2$  = age (years),  $Y$  = price (lakhs)

Number of Samples: 6    Number of Features: 2

Target / Class: Selling price in lakhs

Description: Area=[1.0–3.5]; Age=[10–2]; Price=[20–43].

## PROGRAM

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

X1 = np.array([1.0, 1.5, 2.0, 2.5, 3.0, 3.5])
X2 = np.array([10, 8, 6, 5, 3, 2], dtype=float)
Y = np.array([20, 24, 29, 32, 38, 43], dtype=float)

Xb = np.c_[np.ones(len(X1)), X1, X2]
t = np.linalg.inv(Xb.T @ Xb) @ Xb.T @ Y
b0, b1, b2 = t
print(f"Plane: Y = {b0:.4f} + {b1:.4f}*Area + {b2:.4f}*Age")

p = Xb @ t
mse = float(((Y - p) ** 2).mean())
r2 = float(1 - ((Y - p) ** 2).sum() / ((Y - Y.mean()) ** 2).sum())
print(f"MLR: MSE={mse:.4f} R2={r2:.4f} RMSE={np.sqrt(mse):.4f}")

new = b0 + b1 * 2.8 + b2 * 4
print(f"Predicted price (Area 2.8, Age 4) = {new:.2f} lakhs")

fig = plt.figure(figsize=(7, 5))
ax = fig.add_subplot(111, projection="3d")
ax.scatter(X1, X2, Y, s=40, label="Houses")
a = np.linspace(0.8, 3.7, 20)
g = np.linspace(1, 11, 20)
A, G = np.meshgrid(a, g)
ax.plot_surface(A, G, b0 + b1 * A + b2 * G, alpha=0.4)
ax.scatter([2.8], [4], [new], s=80, marker="*",
           label="Prediction (2.8, 4)")
ax.set_xlabel("Area X1")
ax.set_ylabel("Age X2")
ax.set_zlabel("Price Y (lakhs)")
ax.set_title("Multiple Linear Regression: Price vs Area and Age")
ax.legend()
plt.tight_layout()
plt.savefig("exp6_plane.png", dpi=150)
```

## OUTPUT

Plane:  $Y = 10.4286 + 9.1429 \cdot \text{Area} + 0.0000 \cdot \text{Age}$

MLR: MSE=0.3810 R2=0.9938 RMSE=0.6172

Predicted price (Area 2.8, Age 4) = 36.03 lakhs

Multiple Linear Regression: Price vs Area and Age

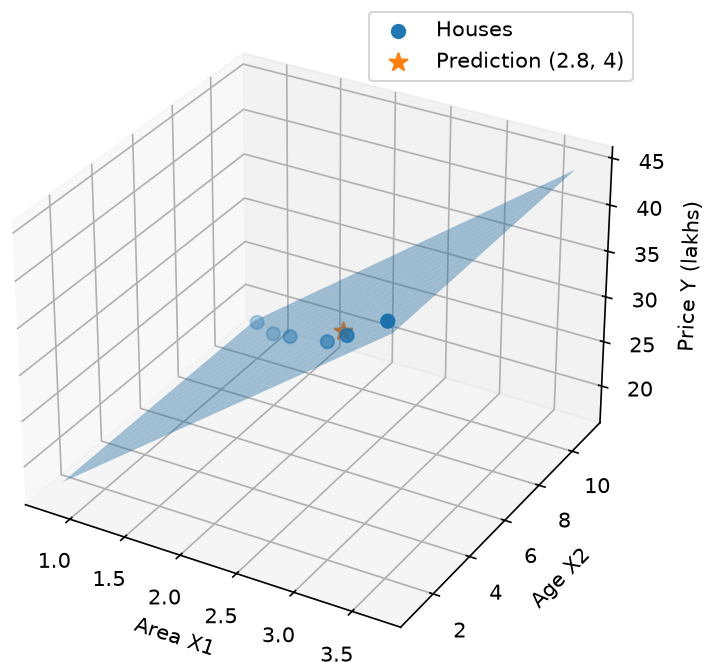


Fig. 1: Houses, fitted plane and predicted house (star).

## RESULT

The above program was implemented and executed successfully using Python, and the corresponding output was obtained and verified. The results of the **multiple linear regression** were analyzed and found to be satisfactory, thereby achieving the desired objective of the experiment.

Faculty Signature: \_\_\_\_\_

Date: \_\_\_\_\_